

Tool 3 – Updates – Sniff53

The feedback that I received from the reviews included the following:

1. Clearer reporting when malformed text log input is skipped or ignored.
2. Capture real DNS traffic to test the tool.
3. Save the output from the tool to a file that can be saved.

The tool has now been updated to parse the input file and catches four types of issues. They include:

Skipped lines

Too few fields (missing query type column)

Invalid source IPs (octet out of range)

Invalid domain names (those lacking a dot)

```
patrickturner@PDTUbuntu:~/Desktop/tool3$ python3 sniff54.py testlog.log

Sniff53
source: testlog.log | threshold: 0.4 | window: 60s

[!] 4 line(s) had unparseable timestamps – used current time as fallback:
line 7: '2024-01-1sdfsdfsfdfsfdfsfdf5T10:00:06' → used 09:38:05
line 35: 'vvvvvvvvvvvvvvvvvvvv2024-01-15T10:02:02' → used 09:38:05
line 36: '2024-01-15T10:02:03vvvvvv' → used 09:38:05
line 37: 'asdfsfsdf2024-01-15T10:02:04' → used 09:38:05

[!] 3 line(s) skipped due to formatting errors:
line 19: invalid source IP: '192.168.99999.15'
       '2024-01-15T10:00:33 192.168.99999.15 www.reddit.com A NOERRO...'
line 24: invalid source IP: '192.168.555.14'
       '2024-01-15T10:00:48 192.168.555.14 outlook.office365.com A N...'
line 36: invalid source IP: '10.0.9999990.99'
       '2024-01-15T10:02:03vvvvvv 10.0.9999990.99 ZXhmaWx0cmF0ZWQgb3...'
```

Capture live DNS traffic:

I took three captures (pcap1, pcap2, and pcap3 which I have included in the repository. The tool hit on a few items and it looks like my laptop is reaching out for Microsoft servers as it was mostly A and AAAA records. I believe it is difficult to capture a real DNS tunneling issue going on from a cable modem. This is a screenshot of pcap1 with a min-score of 0.4.

```

patrickturner@PDTUbuntu:~/Desktop/tool3$ python3 sniff53v2.py --min-score 0.4 pcap1.pcap

Sniff53
source: pcap1.pcap | threshold: 0.4 | window: 60s

▶ 68.199.115.157 4 flagged queries, peak 0.435
  reason: entropy, host_pattern, query_rate

19:08:20 0.435 LOW A mobile.events.data.microsoft.com
19:08:20 0.435 LOW AAAA mobile.events.data.microsoft.com
19:08:21 0.435 LOW A mobile.events.data.microsoft.com
19:08:21 0.435 LOW AAAA mobile.events.data.microsoft.com

— Summary —
Queries analysed : 68
Flagged          : 4 (5.9%)

HOST          HITS  PEAK  REASON
68.199.115.157 4    0.43  entropy, host_pattern, query_rate

patrickturner@PDTUbuntu:~/Desktop/tool3$

```

Save the output from the tool to a file.

Updated the tool to add a logging option to the command. The command line now accepts `-log <logfile>` and writes all information on screen to file.

Examples:

```
Python sniff53v2.py capture.pcap --log logfile.txt
```

```
Python sniff53v2.py queries.log --log logfile.txt
```

```
Python sniff53v2.py capture.pcap -- min-score 0.5 --log minscore05.txt
```

```
Python sniff53v2.py capture.pcap --features --log features.txt
```

```
Python sniff53v2.py capture.pcap --summary --log summary.txt
```

```

patrickturner@PDTUbuntu:~/Desktop/tool3$ python3 sniff53v2.py testlog.log --summary --log summary.txt

[!] 4 line(s) had unparseable timestamps - used current time as fallback:
line 7: '2024-01-1sdfsdfsdffsdffsd5T10:00:06' -> used 11:24:53
line 35: 'vvvvvvvvvvvvvvvvvvvv2024-01-15T10:02:02' -> used 11:24:53
line 36: '2024-01-15T10:02:03vvvvvv' -> used 11:24:53
line 37: 'asdfsfsdf2024-01-15T10:02:04' -> used 11:24:53

[!] 3 line(s) skipped due to formatting errors:
line 19: invalid source IP: '192.168.99999.15'
'2024-01-15T10:00:33 192.168.99999.15 www.reddit.com A NOERRO...'
line 24: invalid source IP: '192.168.555.14'
'2024-01-15T10:00:48 192.168.555.14 outlook.office365.com A N...'
line 36: invalid source IP: '10.0.9999990.99'
'2024-01-15T10:02:03vvvvvv 10.0.9999990.99 ZXhmaWx0cmF0ZWQgb3...'

Summary
-----
Queries analysed : 87
Flagged          : 27 (31.0%)

HOST          HITS   PEAK   REASON
10.0.0.99      6      0.66   entropy, host_pattern, long_subdomain, query_rate, txt_type
10.0.0.55      5      0.55   digit_ratio, entropy, host_pattern, query_rate
10.0.0.77      8      0.55   entropy, host_pattern, nxdomain, query_rate
10.0.0.88      8      0.47   entropy, host_pattern, query_rate, txt_type

results written to: summary.txt
patrickturner@PDTUbuntu:~/Desktop/tool3$

```

I implemented all the feedback from the reviews.

I have learned several lessons in developing this tool. The first lesson is the complexity of Python and realizing I am just touching the surface of what Python can do. The second lesson is the intricacies of Claude AI. AI is only as good as what you prompt it to do. If you have weak prompts or inconsistent goals – you will get a weak response. The #1 learning experience is from my peers. I have learned an incredible amount from feedback as well as learning new technologies by reviewing peers' projects.

If I were to continue with this project, I would turn it into a live capture with an alerting system so that it could be implemented in a SOC.